

Designing Experiments

1) SAT Coaching Program

- a) The problem with this experiment is that there's no control group to compare against, and it would be very difficult to acquire a control group since there are many factors that could affect an individual's personal ability to study. Measuring only the improvement in the students who attended the SAT coaching program doesn't isolate for the effects of the program and fails to identify other confound variables.
- b) Some alternatives to why there was such a drastic improvement in SAT scores could be that the students who want to find success in the SAT would already seek out classes/additional resources to help them study; these people are already on a path to improvement, thus creating a biased test group. Another reason could be that it's simply their second time taking the test, and having already taken the test before, they now have that experience and understanding for the format it is in.
- c) Since the scenario indicates that the students chosen for the "random" sample were students who attended the program, this would be an observational study. Had the study first taken students who had no awareness of the coaching program and no inclination to take an SAT coaching program, and then introduced them and had them participate, then that would be an experiment.

2) Therapeutic Benefits from Pets

- a) This study is observational because the researchers selected people who already have pets, they did not force pets onto heart attack patients.
- b) Yes, the comparison group that the pet owners were compared against were non-pet owners who also suffered a heart attack.
- c) Yes, there are almost always other alternative explanations or confounding variables. What immediately comes to mind is that the owners of pets may involve themselves in more passive exercise, such as walking their dog, which makes them more likely to be living a healthier lifestyle. There's also the socioeconomic factor, that owning a pet costs money and being able to afford that lifestyle means having greater access to ways of benefiting their health.

3) Vitamin C v. Common Cold

- a) This study is experimental, since they formed two separate groups and only introduced a new factor, one that wasn't already present, to one of the groups.

- b) Yes, because they separated the subjects into two distinct groups, one of which was given only one differentiating factor from the other, they have both the control group and the treatment group.
 - c) Yes, they made sure they had a control group to compare against, and chose that control group based on randomness, rather than any other factors.
 - d) Yes, the study was randomized by way of a coin flip, which determined which group each subject would end up in, ultimately resulting in whether they received the treatment or not.
 - e) Some confounding factors that could influence the likelihood of someone catching a cold, such as diet, age, or stress from other parts of their life. Since test subjects were randomly selected, given a large enough test group, these factors should be randomly distributed between the groups equally, though these biases do always need to be checked afterward in case. There is also the issue of the placebo effect, and that individuals who were receiving the treatment might've felt like they should be healthier since they were receiving treatment, causing them to be less likely to catch a cold. As described in the article, a double blind could be used here to minimize placebo and biases.
- 4) Fluoridation and Tooth Decay
- a) Consider the variables
 - i) The independent variable is fluoridated water, against non fluoridated water
 - ii) The dependent variable is tooth decay, which can be observed by an orthodontist/dentist/dental professional
 - b) Testable hypothesis
 - i) The null hypothesis is that fluoridated water has no effect on tooth decay in college students, meaning that students with fluoridated water develop just as many cavities as those who don't
 - ii) The alternative hypothesis is that fluoridated water reduces tooth decay in college students, meaning that students with fluoridated water develop fewer cavities than those who don't.
 - c) Experimental treatments that manipulate the independent variable
 - i) In order to minimize other variables, the experiment would have to supply all test participants with all the drinking water they would be consuming during the duration of the experiment. The treatment group would receive

enough fluoridated drinking water for the entire duration of the treatment, with a specific amount of fluoride in it that could be controlled, whereas the control group would receive enough un-fluoridated water for the duration of the experiment. This way, the only factors regarding the drinking water should be the amount of fluoride in it.

d) Assign subjects to groups

- i) Ideally, we would need to find a college campus whose community lacks fluoridated water so that there's no accidental exposure to it, and then screen the participants for any preexisting tooth decay. Then, we would randomly assign each participant to one of the groups and supply them with their drinking water. Obviously, there are other confounding issues existing, such as their diet or lifestyle, but with a large enough sample size, there should be enough to offset those biases.

e) How to measure the dependent variable

- i) Tooth decay would need to be measured by a professional, someone who is able to categorize and quantize the "amount" of tooth decay an individual has. This would be done both before, during, and after the experiment so that we could have an accurate measure of how quickly tooth decay accumulated in all of the test subjects, and compare the groups against each other.